

Task 2: Client Profitability & Spread Recommendation

Expected PnL per trade at each horizon (eq. 5), the aggregate expected PnL per trade with uniform closing weights (eq. 6, `agg_pnl`), and the break-even minimum half-spread d^* are shown below. All quantities are sample averages from the LP's (our) perspective.

client	tau=5	tau=10	tau=15	tau=20	tau=25	tau=30	agg_pnl	delta_star
A	1.268232	1.540175	1.737274	2.061331	2.292701	2.485176	1.897481	0
B	1.211394	1.342476	1.542005	1.725621	1.909079	2.058551	1.631521	0
C	1.07672	1.164963	1.189768	1.162116	1.2285	1.216464	1.173089	0
D	0.867836	0.681538	0.492889	0.177152	-0.07606	-0.3469	0.299409	0.009614
E	0.652815	0.302584	-0.10807	-0.57245	-1.06874	-1.61435	-0.40137	0.018983
F	0.445223	-0.17643	-0.91678	-1.75096	-2.61386	-3.6294	-1.44037	0.032866

Classification (based on sign of the aggregate from equation 6)

Net profitable clients (for the LP): A, B, C and D

Net costly clients (for the LP): E and F

A, B and C are profitable at every horizon. D is profitable in aggregate (+0.30) but turns negative beyond $\tau=20$, so it is marginal. E (-0.40) and F (-1.44) are net-costly, their positive near-term edge is overwhelmed by losses at longer horizons, exactly what was expected from their steep adversity profiles in Task 1.

Minimum half-spread δ^* and its relation to adversity

Quoting at $M_0 \pm \delta$ makes the effective trade price (TP') as:

$$TP' = M_0 \pm \text{side} \cdot \delta$$

So the aggregate PnL becomes $= \text{side} \cdot V \cdot (\bar{M} - M_0) + V \cdot \delta$

Setting its average equal to zero (and adding negative clipping) gives `delta_star` as:

$$\delta^* = \max(0.0, \frac{-E[\text{side} \cdot V \cdot (\bar{M} - M_0)]}{E[V]})$$

(Note: negative clipping was added because bid-ask spread can never be negative)

Intuitively, `delta_star` is the per unit adverse selection cost the spread must recover.

The recommended δ^* tracks the task-1 adversity ranking almost perfectly with A, B, C and D having negative δ^* (which was clipped to zero because bid-ask spread cannot be zero). This means that A, B and C are profitable even at mid, so no protective spread is needed. But D, E and F require progressively wider half-spreads δ^* (as you can see from the table). The steeper a client's adversity profile, the larger will be the half-spread we (the LP) need to charge to break even. Adversity and δ^* are different views of the same toxicity.